

Structural design and scantlings

SDO/STR/SCANT-021 Rev B

RESTRICTED

5.1 Still water bending moment

The hull girder section modulus at amidships shall be calculated for both hogging and sagging conditions, combining the still water bending moment from the worst realistic loading condition with the wave bending moment derived from the applicable classification rule. A minimum margin of 10 percent above the rule-required section modulus shall be retained at the initial design stage to absorb weight growth over the service life.

5.4 Shell plating

Transverse frame spacing shall be 600 mm throughout the machinery spaces and 700 mm elsewhere unless the arrangement requires otherwise. Bottom shell plating thickness in way of the machinery spaces shall not be less than 9 mm in grade AH36 steel. Where plating thickness changes between adjacent strakes, the difference shall not exceed 3 mm and the taper shall extend over not less than three frame spaces.

9.2 Shock grades

Equipment essential to propulsion, steering, firefighting and combat capability shall be qualified to Grade A shock. Grade B applies to equipment whose failure would not immediately impair fighting capability but could injure personnel or damage Grade A items. Deck-mounted Grade A equipment shall be qualified by hammer or barge test, and mounting seats shall be shown to transmit no amplification greater than 1.5 in the 5 to 100 Hz band.